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Weather Trend Forecasting

Tech Assessment

Prepared for:
PM Accelerator

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By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders. This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, surround you with the right PM ecosystem, and discover the new world of AI product management skills.

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1 Executive Summary

This report summarizes an end-to-end weather analytics workflow built on the Global Weather Repository dataset. The dataset contained data spanning over 200 countries from 2024-05-16 to 2026-04-18. Data quality was strong overall: there were no duplicate rows removed, no missing-value issue highlighted in the notebook output, and only 2 dates were removed to exclude from the modelling workflow.

Anomaly detection flagged some observations using a combined Isolation Forest and temperature Z-score rule. In forecasting, the stacked ensemble delivered the strongest performance with an MAE of 0.613, outperforming Random Forest (1.074), XGBoost (1.220), and ARIMA (4.044)

2 Methodology

The source data is stored in `GlobalWeatherRepository.csv`. It includes timestamped observations together with weather, air-quality, and location fields such as temperature, wind, pressure, humidity, cloud cover, precipitation, visibility, UV index, and pollutant measurements.

The cleaning and preparation workflow followed the notebook logic closely:

- Parsed `last_updated` as a datetime field.
- Checked for missing values and duplicate records.
- Added calendar features: year, month, day, day of week, and day of year.
- Created cyclical encodings `doy_sin` and `doy_cos` to help models capture annual seasonality.
- Removed observations from two explicit dates used as exclusions in the notebook logic.
- Resampled to daily frequency, forward-filled gaps, and created 14 lag features for temperature forecasting.

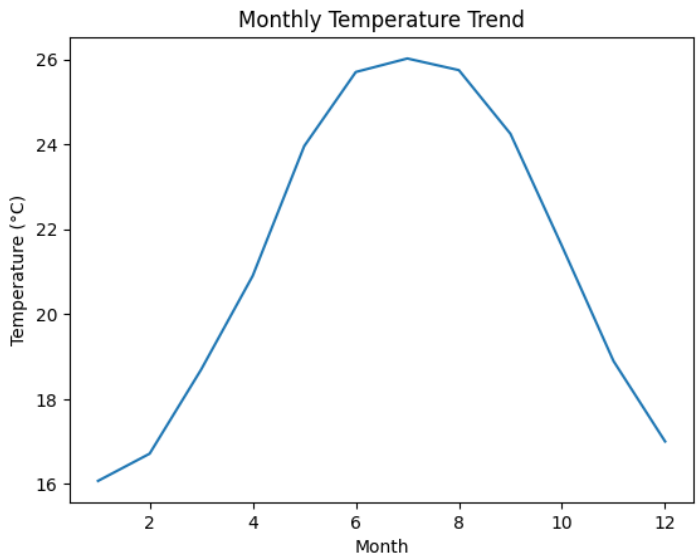

Figure 1: Average temperature by calendar month, showing the strongest warming in mid-year.

The analytical approach combined descriptive analysis with predictive modelling. Isolation Forest and Z-score anomaly detection were used to identify unusual weather states. Forecasting performance was then compared across Random Forest, ARIMA, and XGBoost. Finally, the three forecasts were blended with a linear meta-model to create a stacked ensemble.

3 Analysis

The exploratory analysis shows that the data contains both strong seasonal structure and broad geographic variation. Monthly temperature averages rise into the middle of the year and cool into the first quarter, which is consistent with global seasonal patterns in an internationally mixed dataset.

The correlation view suggests that most individual weather and air-quality variables have only modest linear relationships with temperature in the full global sample. This matters because it helps explain why non-linear models outperform ARIMA: the data mixes many geographies, climates, and local effects that are unlikely to be represented well by a simple linear time-series assumption alone.

Anomaly detection added an engineered feature that marks unusual observations. The global temperature series itself is relatively smooth after daily aggregation, although regional views show that different locations move on distinct trajectories.

Additional analysis linked weather conditions to air-quality behavior. The linear correlation between temperature and PM2.5 was weak at 0.055, which indicates that temperature alone is not a strong predictor of particulate pollution at a global scale. Spatial analysis reinforces this point: average temperature

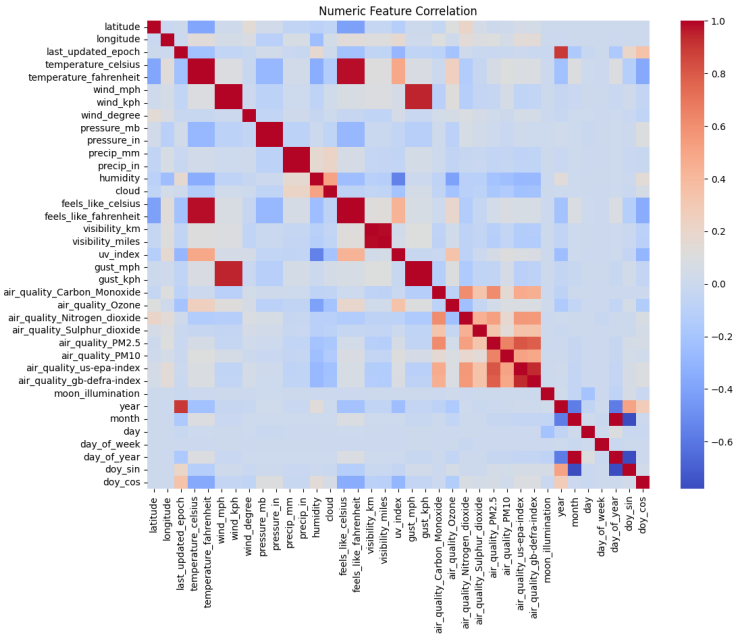

Figure 2: Correlation heatmap for a core subset of weather and air-quality features.

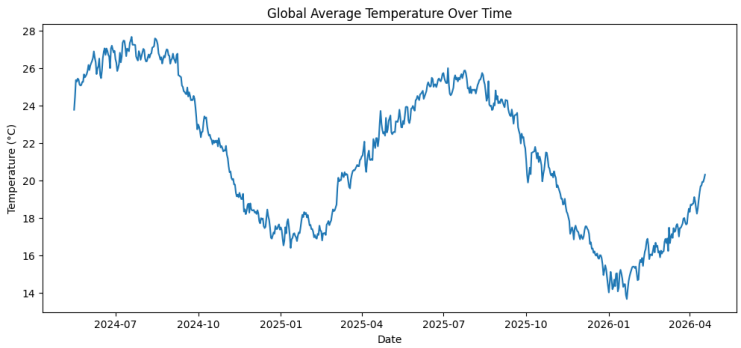

Figure 3: Daily global average temperature after resampling and forward-filling.

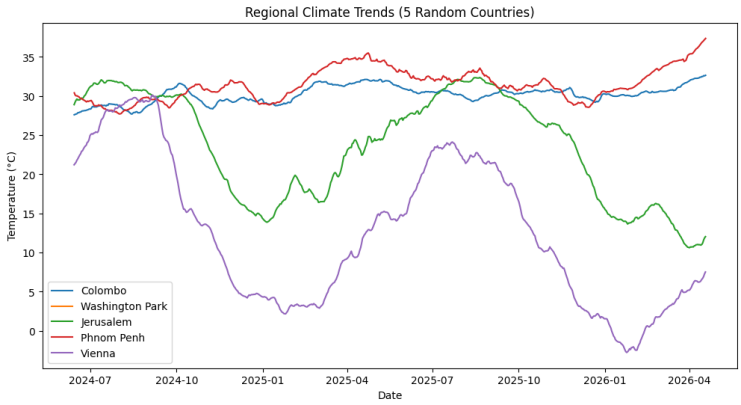

Figure 4: Thirty-day rolling regional trends for five sampled locations.

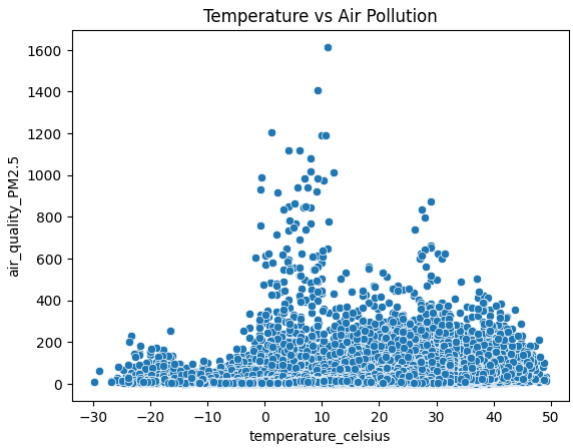

Figure 5: Weak global relationship between temperature and PM2.5 concentration.

differs strongly across geographies, with Saudi Arabien as the hottest country in the sample at 45.00°C and Mongolia as the coolest at 3.50°C.

4 Results

The modelling dataset produced an 80/20 split for training and testing after lag creation and the time-based split. Forecast evaluation used mean absolute error so that model comparison stayed directly interpretable in degrees Celsius.

The results show a clear ranking. ARIMA lagged well behind the machine-learning approaches, suggesting that temperature movement in this dataset depends on richer multivariate context than a simple univariate time-series specification can capture. Random Forest and XGBoost both performed well, with Random Forest slightly ahead. The ensemble delivered the lowest error overall, cutting MAE by roughly 43% versus the best single base model.

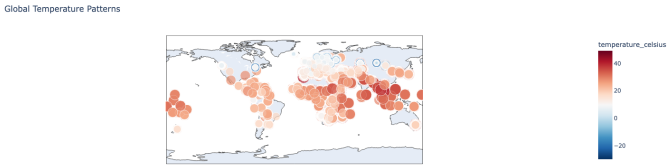

Figure 6: Geographic spread of sampled temperatures by latitude and longitude.

Table 1: Forecast model evaluation

Model	MAE
Random Forest	1.074
ARIMA	4.044
XGBoost	1.220
Stacked Ensemble	**0.613**

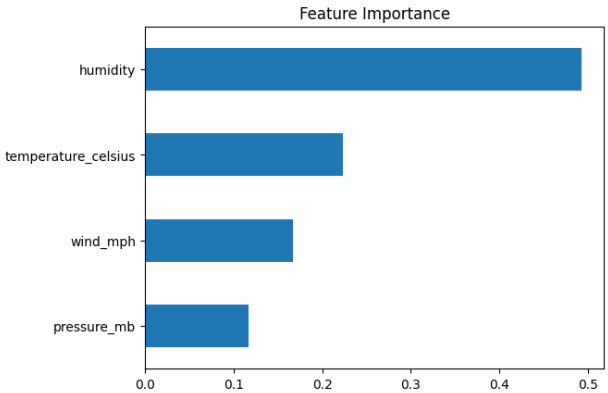

Figure 7: Random Forest feature importance for the PM2.5 side analysis.

Feature importance analysis on the environmental side shows that temperature, humidity, wind, and pressure all contribute to PM2.5 modelling, but their relative impact is uneven. This supports the project's broader conclusion that multivariate relationships are present, though not strongly linear.

5 Conclusion

This project demonstrates a complete workflow from raw weather observations to forecast-ready features, benchmarked models, and decision-oriented insights. The data cleaning and EDA stages showed a strong seasonal pattern and wide geographic diversity. The advanced analysis added anomaly, environmental, and spatial context that improved interpretation of the forecasting problem.

The key result is straightforward: a simple stacked ensemble produced the best temperature forecast performance in this workflow. That makes it the strongest candidate for future operationalization, especially if the project

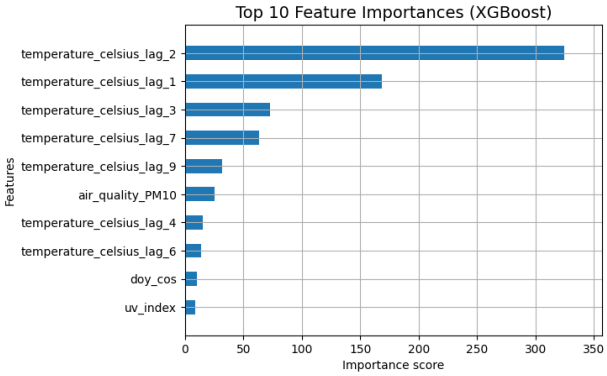

Figure 8: XGBoost feature importance plot for top 10.

expands into location-specific predictions and a more formal model monitoring pipeline.